Fix-round CMSSW emu, file 3: emitted angle distributions, class T vs class C

CMS alpha - IL2-4

CMS alpha - IL1

CMS alpha - ID1-4

CMS beta - IL2-4

CMS beta - IL1

CMS beta - ID1-4

density (unit area)

angle [deg]

class T: TRUE angle

class T: EMITTED

class C linked: EMITTED

class C unlinked: EMITTED

Emitted angle statistics [deg]

group	angle	curve	N	mean	rms	frac_abs_gt60_pct
IL2-4	alpha	class T: TRUE angle	1174067	0.8734	13.13	0.1549
IL2-4	alpha	class T: EMITTED	1174067	0.4702	14.48	0.2318
IL2-4	alpha	class C linked: EMITTED	409184	1.522	21.62	1.663
IL2-4	alpha	class C unlinked: EMITTED	231810	2.453	23.19	2.521
IL1	alpha	class T: TRUE angle	719916	2.048	12.87	0.3678
IL1	alpha	class T: EMITTED	719916	1.633	14.12	0.4851
IL1	alpha	class C linked: EMITTED	83606	1.927	21.99	3.393
IL1	alpha	class C unlinked: EMITTED	95569	2.064	20.04	2.276
ID1-4	alpha	class T: TRUE angle	3229453	0.01008	5.199	0.09624
ID1-4	alpha	class T: EMITTED	3229453	0.3547	8.145	0.1416
ID1-4	alpha	class C linked: EMITTED	476705	0.3621	19.01	2.715
ID1-4	alpha	class C unlinked: EMITTED	826596	0.4069	14.2	1.014
IL2-4	beta	class T: TRUE angle	1174067	-0.0431	43.83	17.85
IL2-4	beta	class T: EMITTED	1174067	-0.7843	43.79	18.29
IL2-4	beta	class C linked: EMITTED	409184	-0.4785	38.1	10.9
IL2-4	beta	class C unlinked: EMITTED	231810	-0.4387	42.74	17.84
IL1	beta	class T: TRUE angle	719916	0.04599	59.68	52.97
IL1	beta	class T: EMITTED	719916	-0.3922	59.52	52.59
IL1	beta	class C linked: EMITTED	83606	-0.01315	59.7	51.83
IL1	beta	class C unlinked: EMITTED	95569	-0.3917	61.42	56.37
ID1-4	beta	class T: TRUE angle	3229453	0.005194	14.37	0.1395
ID1-4	beta	class T: EMITTED	3229453	1.339	21.47	0.2307
ID1-4	beta	class C linked: EMITTED	476705	1.128	28.99	4.855
ID1-4	beta	class C unlinked: EMITTED	826596	1.126	23.79	1.264

Saturation (clamp at +-(90 - 0.01) deg) and |cot| > 12 rates [%], per group x class

group	cls	n	satAlpha_pct	satBeta_pct	frac_abs_cot_gt12_alpha_pct	frac_abs_cot_gt12_beta_pct
IL2-4	T	1174067	0.00264	0.02138	0.01831	0.04088
IL2-4	C	640994	0.03292	0.06053	0.253	0.2056
IL1	T	719916	0.0009723	0.03653	0.04917	0.1413
IL1	C	179175	0.03795	0.2216	0.4861	2.903
ID1-4	T	3229453	0.0001858	0.06617	0.006812	0.07596
ID1-4	C	1303301	0.004757	0.1307	0.1582	0.3773